blockchain, which leads to the first line of defense supporting privacy, the three major models for blockchain networks: public, private, and permissioned.

12.7 Public, private, and permissioned networks

Bitcoin (also called the permissionless blockchain) is a working example of a public blockchain. The primary purpose of the Bitcoin blockchain is to support a decentralized, peer-to-peer payment system. It’s meant to be a transparent, permissionless *public* system that anyone can join and leave as they wish, as in any other bearer payment system, such as transacting cash. If you pay cash for an item you purchase in a store, for example, nobody asks you for your signature or needs you to authorize the payment. Similarly, Bitcoin enables a peer-to-peer digital payment system without any intermediaries.

When the use cases for blockchain expanded beyond the simple payment system into business areas such as personal health care systems and financial systems, privacy and restricted access became necessary. Even in the public payment systems, it became apparent that the whole chain may not be relevant to all the participants. Business transactions in a Buffalo school district may not be relevant to the Nairobi tourism board, for example. Such thoughts and ideas resulted in the creation of permissioned blockchain (figure 12.5), in which only permitted participants can transact and take part in the blockchain operations.

The permissioned blockchain is also known as a consortium blockchain based on its everyday use cases in specific vertical